Equation Sample 4

Dataset-expansion sample

1 Derivation

The following display is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

$$\begin{aligned}
||f||_{1_\mu} + ||g||_{1_\mu} &= \int\limits_{x\in\mathcal{D}} |f(x)| + |g(x)| d\mu(x) \\
&= + \int\limits_{\substack{x\in\mathcal{D}\\ f(x)\geq 0, g(x)\geq 0,}} f(x) + g(x) d\mu(x) - \int\limits_{\substack{x\in\mathcal{D}\\ f(x)<0, g(x)<0,}} f(x) + g(x) d\mu(x) \\
&\quad + \int\limits_{\substack{x\in\mathcal{D}\\ f(x)\geq 0, g(x)<0,}} f(x) - g(x) d\mu(x) + \int\limits_{\substack{x\in\mathcal{D}\\ f(x)<0, g(x)\geq 0,}} g(x) - f(x) d\mu(x) \\
&= + \int\limits_{\substack{x\in\mathcal{D}\\ f(x)\geq 0, g(x)\geq 0,}} \max(f(x), g(x)) + \min(f(x), g(x)) d\mu(x) \\
&\quad - \int\limits_{\substack{x\in\mathcal{D}\\ f(x)<0, g(x)<0,}} \max(f(x), g(x)) + \min(f(x), g(x)) d\mu(x) \\
&\quad + \int\limits_{\substack{x\in\mathcal{D}\\ f(x)\geq 0, g(x)<0,}} \max(f(x), g(x)) - \min(f(x), g(x)) d\mu(x) \\
&\quad + \int\limits_{\substack{x\in\mathcal{D}\\ f(x)<0, g(x)\geq 0,}} \max(f(x), g(x)) - \min(f(x), g(x)) d\mu(x).
\end{aligned}$$